Weekly Review and Practice Practical — VLANs, Trunking, and Router-on-a-Stick

Task 1: On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

A network of three switches was built in Cisco Packet Tracer. Two VLANs were created — VLAN 10 (Sales) and VLAN 20 (Support) — and two PCs were connected to each switch, with one PC assigned to each VLAN based on department.

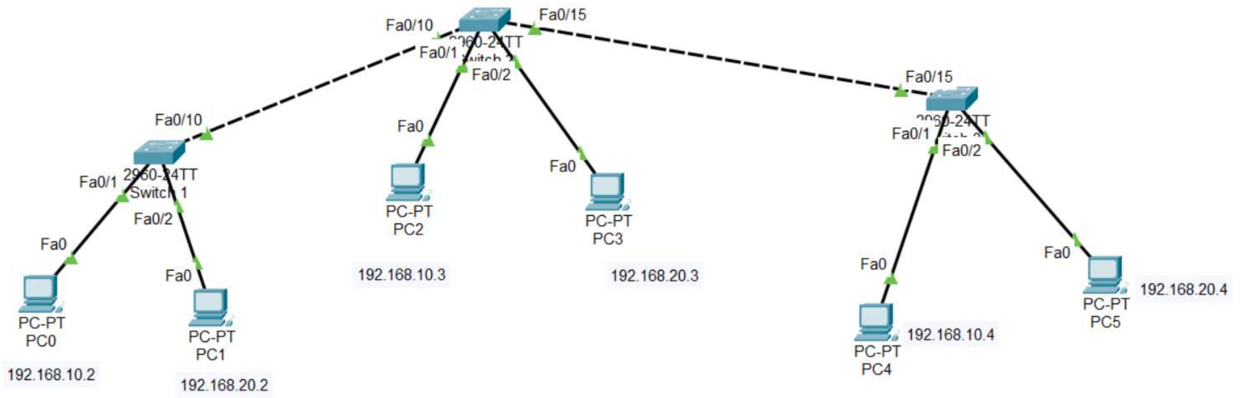


Figure 1: Initial topology — Switch 1, Switch 2, and Switch 3, each with one VLAN 10 PC and one VLAN 20 PC

PC	Switch	VLAN	Department	IP Address
PC0	Switch 1	10	Sales	192.168.10.2
PC1	Switch 1	20	Support	192.168.20.2
PC2	Switch 2	10	Sales	192.168.10.3
PC3	Switch 2	20	Support	192.168.20.3
PC4	Switch 3	10	Sales	192.168.10.4
PC5	Switch 3	20	Support	192.168.20.4

This access port configuration (Fa0/1 → VLAN 10, Fa0/2 → VLAN 20) was applied identically on all three switches.

Task 2: Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

The inter-switch links were configured as 802.1Q trunks carrying both VLANs, so that VLAN traffic could pass between all three switches.

Switch 1 running-config, showing access ports Fa0/1 and Fa0/2 and the trunk configuration on the inter-switch link:

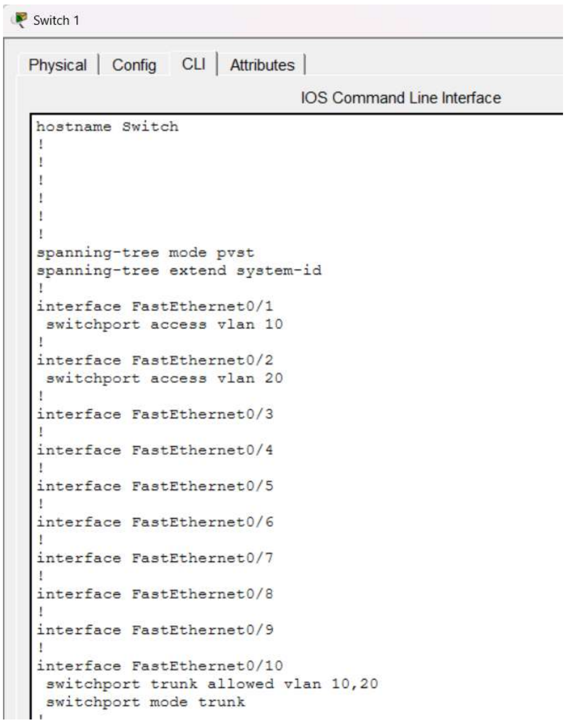


Figure 2: Switch 1 running-config — VLAN access ports and trunk setup

Trunking was verified on all three switches using show interfaces trunk once the router uplink was also added (Task 3):

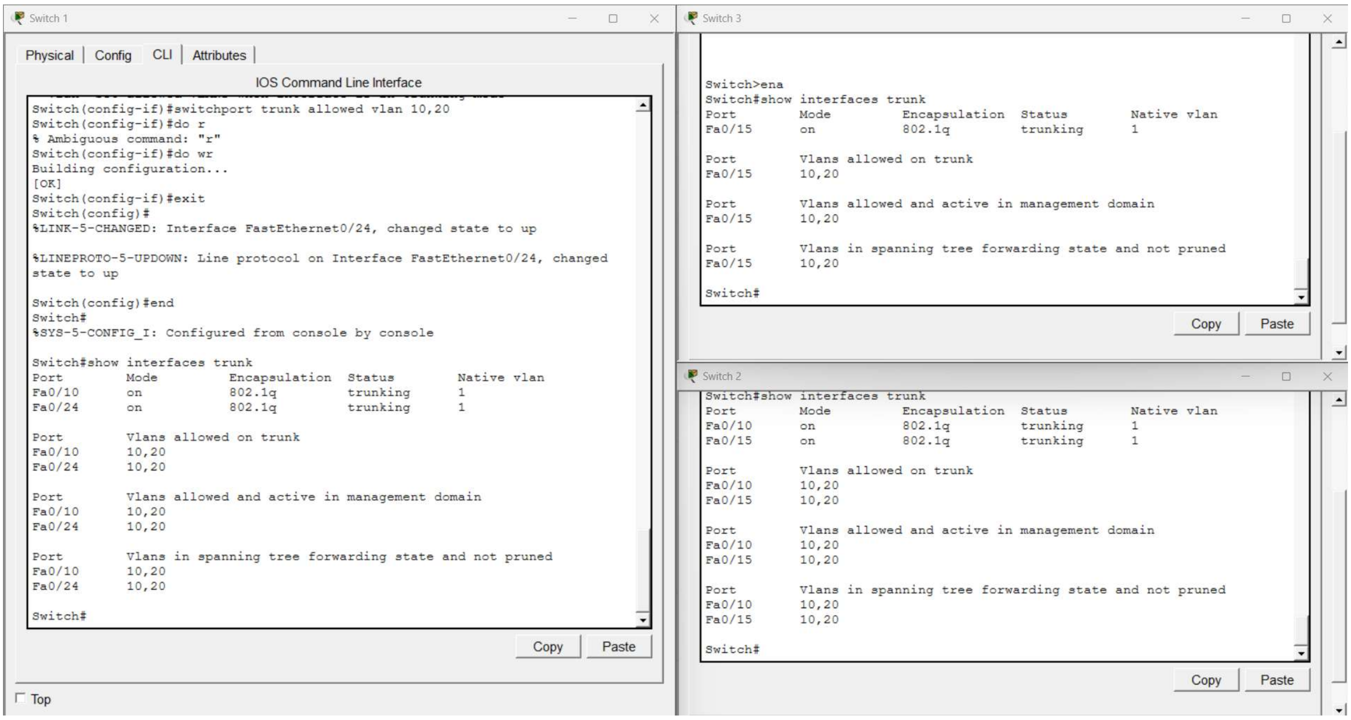


Figure 3: show interfaces trunk output on Switch 1, Switch 2, and Switch 3 — all inter-switch and router-facing links trunking, VLANs 10 and 20 allowed and active

Task 3: Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

A router (2620XM) was connected to Switch 1 to provide Layer 3 routing between VLAN 10 and VLAN 20. The switch port facing the router was configured as a trunk, and two router sub-interfaces were created with 802.1Q encapsulation matching each VLAN.

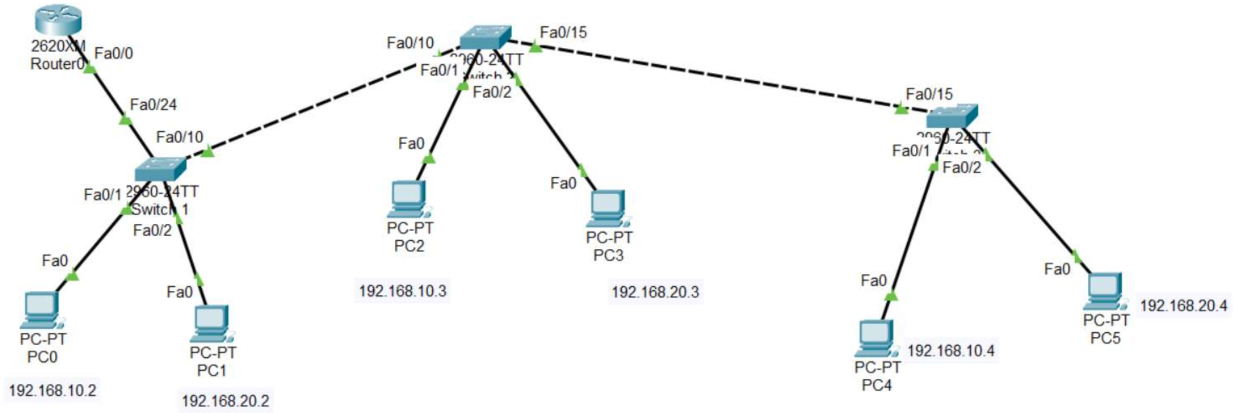
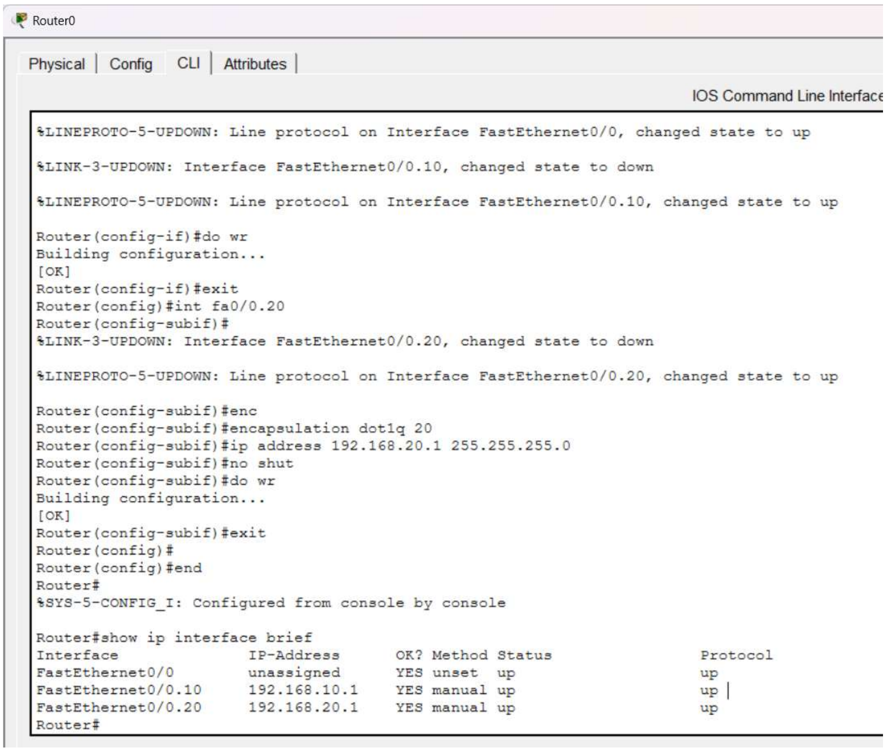


Figure 4: Updated topology with Router0 connected to Switch 1 (Fa0/0 → Switch 1 Fa0/24)

show ip interface brief on the router confirms both sub-interfaces are up/up with the correct gateway addresses, while the parent physical interface Fa0/0 correctly carries no IP of its own:


Router0

Physical | Config | CLI | Attributes |

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to up

Router(config-subif)#enc
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shut
Router(config-subif)#do wr
Building configuration...
[OK]
Router(config-subif)#exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  up                    up
FastEthernet0/0.10     192.168.10.1    YES manual up                    up
FastEthernet0/0.20     192.168.20.1    YES manual up                    up
Router#
```

Figure 5: Router0 sub-interface configuration and show ip interface brief output

Task 4: Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works. Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.

Initial Test — Before Router-on-a-Stick Was Configured

Before Task 3 was completed, PC0 (VLAN 10) could already reach the other VLAN 10 PCs on remote switches, confirming trunking between switches was working correctly for same-VLAN traffic:

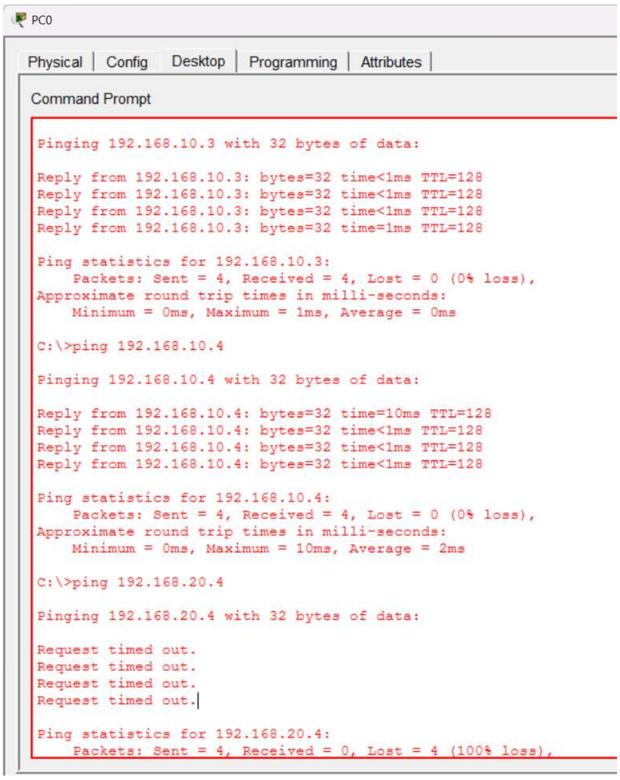


Figure 6: PC0 successfully pings 192.168.10.3 and 192.168.10.4 (VLAN 10, remote switches) — 0% loss

At the same point, a ping from PC0 to 192.168.20.4 (VLAN 20, Switch 3) failed with 100% loss, which was expected since no Layer 3 device existed yet to route between VLANs:

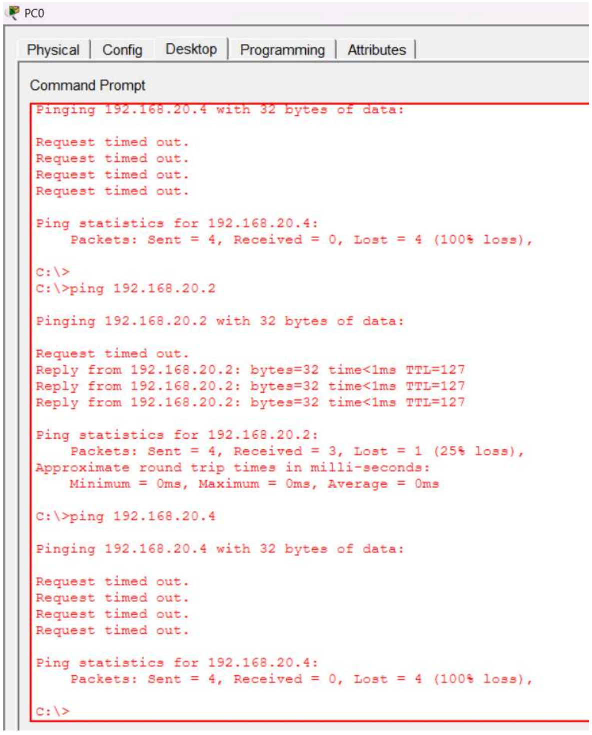


Figure 7: PC0 ping to 192.168.20.4 — 100% loss, prior to inter-VLAN routing setup

Issue Found After Router-on-a-Stick Was Configured

After completing Task 3, PC0 could successfully reach PC1 (192.168.20.2), which is on the same switch as PC0, confirming the router and trunk were correctly routing between VLAN 10 and VLAN 20 locally. However, a ping to 192.168.20.4 (PC5, on Switch 3) still failed with 100% loss.

Diagnosis: since inter-switch trunking was already verified working (Figure 3) and the router sub-interfaces were confirmed up with correct IPs (Figure 5), the fault was on the PC side. PC2, PC3, PC4, and PC5 had not yet been assigned a default gateway. Ping requests were able to reach these PCs, but with no gateway configured, the reply packets had no route back out of the local VLAN — causing every distant-VLAN ping to time out.

Fix Applied

A default gateway was configured on every remaining PC, matching its VLAN's router sub-interface IP:

PC	VLAN	IP Address	Default Gateway
PC1	20	192.168.20.2	192.168.20.1
PC2	10	192.168.10.3	192.168.10.1
PC3	20	192.168.20.3	192.168.20.1
PC4	10	192.168.10.4	192.168.10.1
PC5	20	192.168.20.4	192.168.20.1

Verification

With default gateways correctly assigned on every PC, the ping from PC0 to 192.168.20.4 was re-run and succeeded with 0% packet loss, confirming full inter-VLAN connectivity across all three switches:

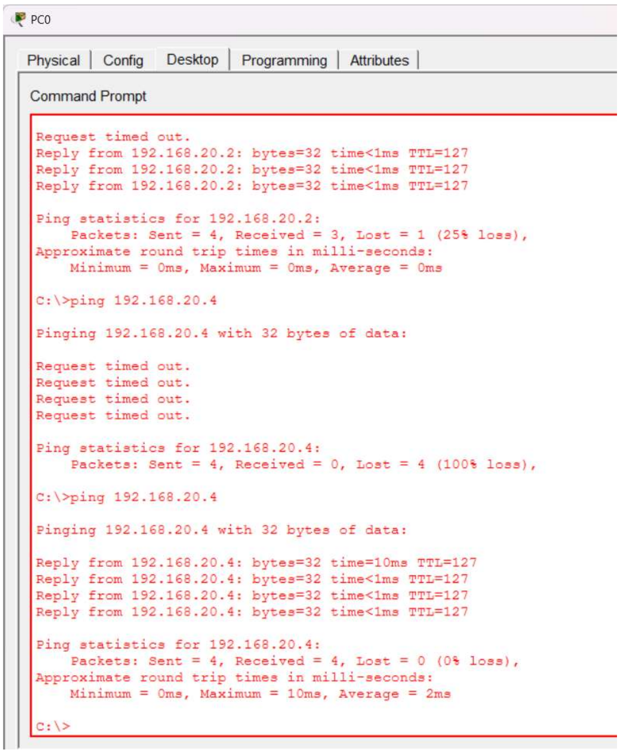


Figure 8: PC0 ping to 192.168.20.4 after assigning default gateways — 0% loss, all replies received

Root cause: missing default gateway addresses on PC2, PC3, PC4, and PC5. Ping requests reached each destination PC across the trunked VLANs, but without a configured gateway the reply traffic could not route back to the originating VLAN. Assigning the correct default gateway (matching each VLAN's router sub-interface IP) resolved the issue and restored full inter-VLAN connectivity.